

Alekya Kudumula

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SUMMARY

Data Engineer with 4 years of experience in building large-scale data pipelines, real-time analytics platforms, and BI solutions across cloud and on-prem environments. Skilled in designing and optimizing ETL/ELT workflows, streaming architectures, and distributed data systems with a focus on performance, scalability, and cost efficiency. Experienced in integrating batch and streaming data from diverse sources into high-performance warehouses and analytics engines like Apache Druid, enabling sub-second query performance for enterprise clients. Adept at collaborating with cross-functional teams to deliver production-ready solutions, automate operational processes, and ensure platform reliability. Strong background in data visualization, predictive modeling, and machine learning applications, supported by hands-on academic and industry projects.

TECHNICAL SKILLS

- ✓ **Programming & Scripting:** Python, SQL, Java, Shell Scripting
- ✓ **Data Engineering & ETL:** Apache Kafka, Apache Druid, Apache Spark, dbt, Airflow, Informatica
- ✓ **Cloud & Storage:** AWS (S3, Redshift, Glue, IAM), GCP (BigQuery, Pub/Sub), Azure Data Services
- ✓ **Databases & Warehousing:** Snowflake, PostgreSQL, MySQL, Oracle, Workday Adaptive
- ✓ **Big Data & Analytics:** Hadoop, MapReduce, Presto, Flink
- ✓ **Data Visualization & BI Tools:** Power BI, Tableau, Looker, Grafana, Metabase
- ✓ **DevOps & Observability:** Docker, Kubernetes, Terraform, Prometheus, ELK Stack
- ✓ **Machine Learning & AI:** Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers, OpenCV
- ✓ **Version Control & Collaboration:** Git, GitHub, Jira, Confluence

EXPERIENCE:

Imply Data, Data Engineer

Aug 2024 - Present

- Designed and built scalable ETL pipelines connecting batch and streaming sources like Kafka, S3, and relational databases to Apache Druid clusters, enabling sub-second query performance.
- Optimized real-time ingestion paths using Druid’s partitioning and indexing techniques to support high cardinality datasets with low-latency analytics.
- Engineered cost-efficient storage strategies combining tiered data management and dynamic compaction—halving infrastructure spend while maintaining query throughput.
- Partnered with client-facing Field Engineering teams to enable seamless deployment and monitoring of Druid/SaaS environments at scale.
- Developed observability tooling to track ingestion health, query performance, and cluster resource utilization—ensuring high availability for enterprise workloads.
- Supported internal and open-source engineering efforts—contributing to Apache Druid’s scalability and robustness, e.g., improving node balancing and query engines for performance.
- Designed data management interfaces in the Polaris Data Management UI to simplify ETL and transformation workflows for streaming and batch data sources.
- Ensured platform reliability via automation of incident response workflows and on-call rotations, collaborating on post-incident reviews to strengthen operational resiliency.
- Collaborated with product and cloud teams to integrate Identity, RBAC, and secure authentication/authorization capabilities across SaaS and on-prem deployments.
- Translated real-time analytics requirements into production-quality, maintainable Data Engineering solutions that enabled customers like Pepsi, Zillow, and Splunk to run interactive dashboards on terabyte-scale datasets.

Berkadia | Data Analyst, India

June 2020 – May 2023

- Designed and optimized ETL/ELT pipelines to extract, transform, and load financial, operational, and sales data from multiple sources into centralized data warehouses.
- Built interactive dashboards and automated visualizations by integrating cleaned datasets into BI platforms such as Power BI, Tableau, and Looker.
- Partnered with stakeholders to translate business requirements into data pipelines, ensuring analytical outputs aligned with organizational goals.
- Modeled data structures to support forecasting, budgeting, and profitability analysis, enabling accurate and timely financial reporting.
- Developed scalable queries and scripts to fulfill ad-hoc and recurring data requests for cross-functional teams, ensuring high performance on large datasets.
- Contributed to data-driven planning cycles by preparing analytical datasets used for annual budget forecasts and executive presentations.
- Automated recurring reporting workflows in Workday Adaptive and other consolidation tools, improving consistency and reducing manual effort.
- Translated complex raw data into structured datasets and provided insights through SQL, Python, and visualization tools for business decision-making.
- Implemented data quality checks, validation processes, and reconciliation routines to maintain integrity and consistency across reporting systems.
- Supported ongoing data initiatives by monitoring pipelines, troubleshooting data anomalies, and enhancing workflows for long-term scalability.

CERTIFICATION: Data Analytics Professional Certificate | Google

EDUCATION

Sacred Heart University, School of Engineering, MS in Computer Engineering, Fairfield, USA GPA3.8/4

December 2024

St. Joseph’s PG and Degree college, Osmania University, B. Com in Computer Science, Hyderabad, INDIA, GPA 9.8/10

June 2020

PROJECTS:

Vision Chat AI Analysis

- Vision Chat is an AI-powered assistant that turns images into interactive conversations, using Vision-Language Models (VLMs) to identify objects, describe scenes, and answer questions.
- Used Stream lit for the interface, Hugging Face Transformers for AI models, OpenCV for image processing, and PyTorch/TensorFlow for deep learning.
- Vision Chat transforms how we engage with visual content, making image analysis intuitive, interactive, and insightful.

Smart Farming Analytics Platform

- Designed a Smart Farming Analytics Platform using IoT sensors to monitor soil moisture, temperature, and humidity for optimized farming.
- Used Arduino for data collection, Python and Scikit-learn for ML-based irrigation scheduling and yield prediction, and Grafana for data visualization. Achieved increase in crop yield and reduction in water usage through data-driven farming insights.